

Superba Tunnel Profiler

How to use — v1.0.4

This app downloads logged temperature data from a Fluke 54-II-B thermometer over its IR cable, and turns it into a time/temperature profile you can view, compare, search, and export.

No installation needed. Extract the SuperbaTunnelProfiler folder anywhere and run **SuperbaTunnelProfiler.exe** inside it — keep the whole folder together, the app stores its data (**superba_profiler.db**) right next to the .exe.

What you need

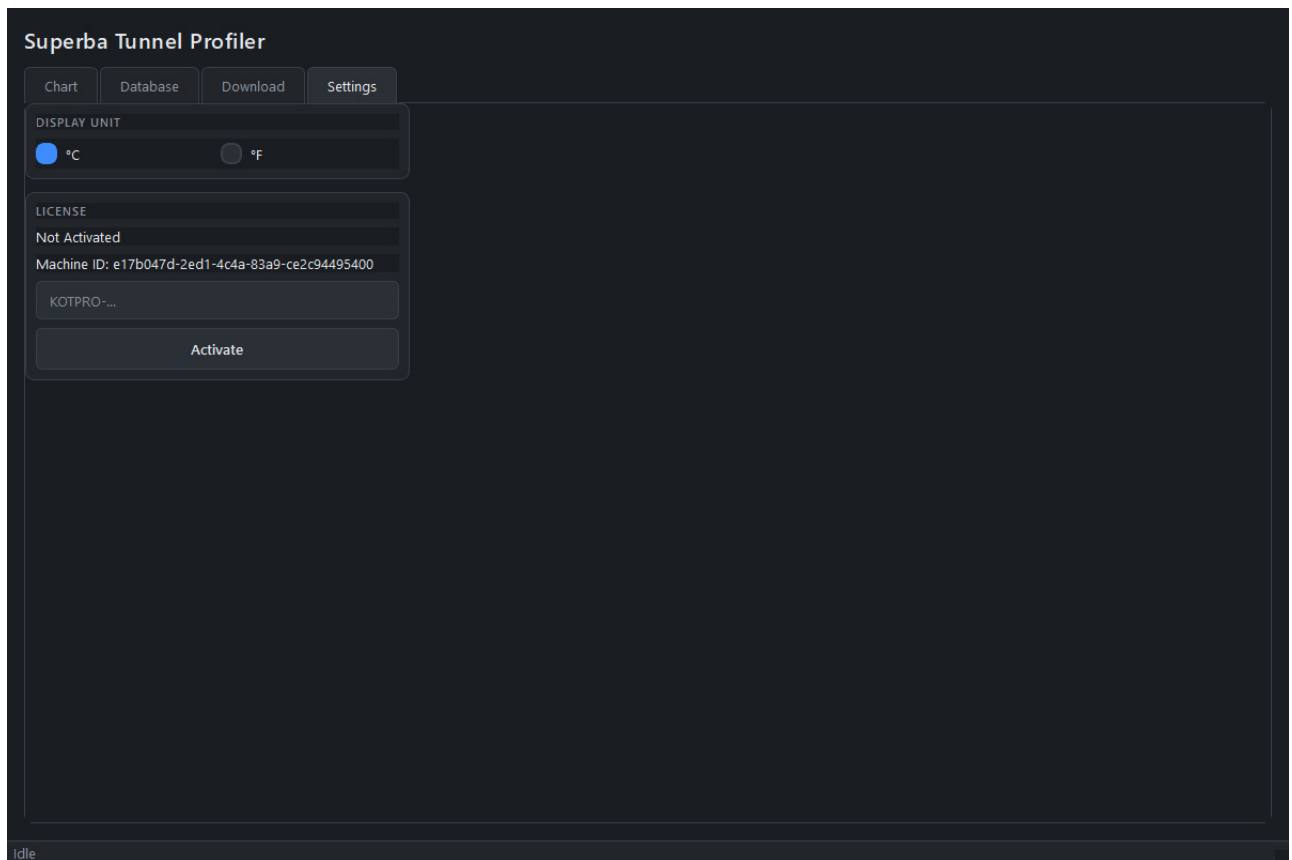
- A Fluke 54-II-B thermometer with logged readings on it
- The Fluke IRUSB Rev.-II cable, connected between the meter and your PC
- Windows — nothing else to install
- An activated license (see below) — required before the app will talk to the meter

The four tabs

Tab	What it's for
Chart	View, compare, and export the profiles you have plotted
Database	Search every profile ever downloaded, and open, export, or print any of them
Download	Pull a new profile off the meter
Settings	Display preferences and license activation

Activating your license

Meter downloads are locked to a license key issued for this specific PC. On the **Settings** tab's License card you'll see this machine's ID and a field to paste a key into:

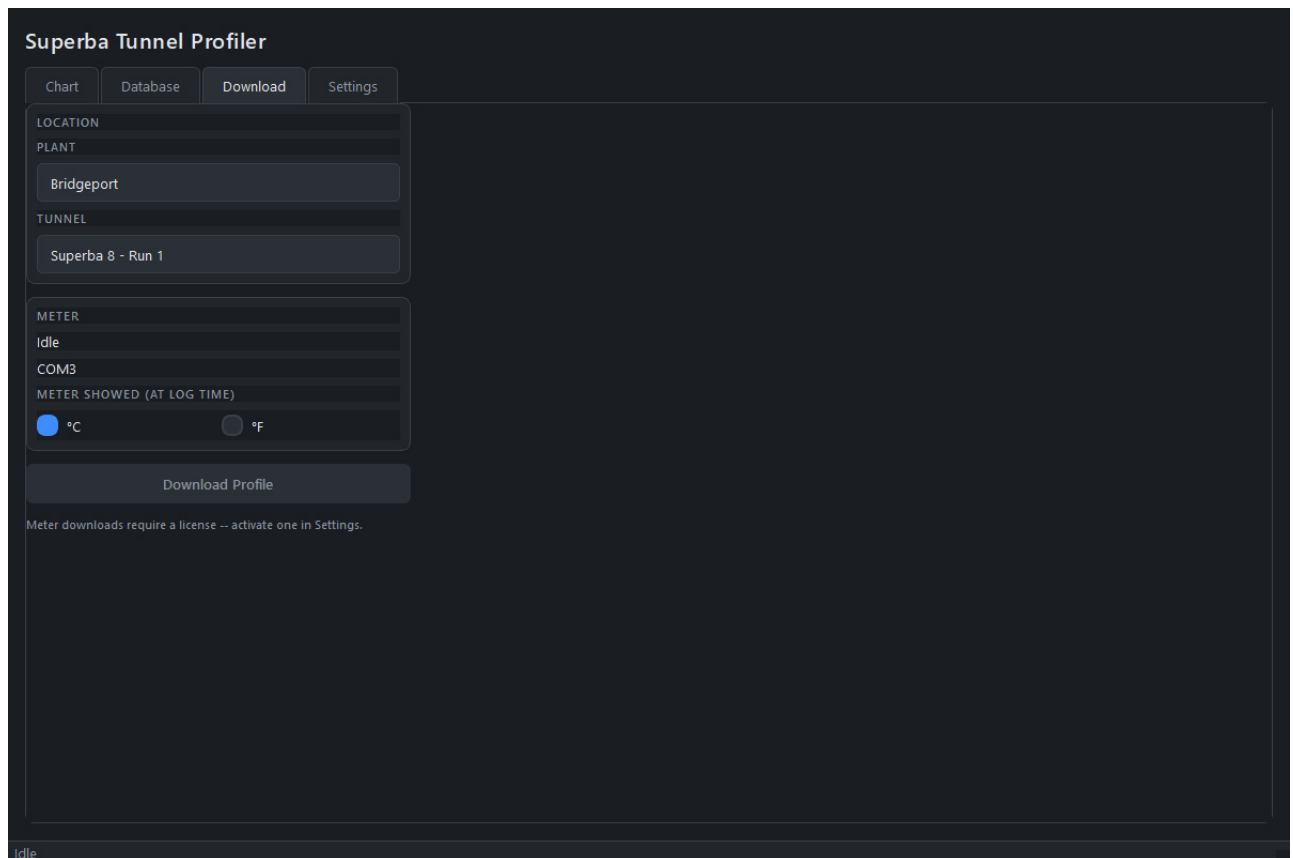


The Settings tab, showing Display Unit and the License card (not yet activated).

Send this machine's ID (shown above) to whoever issues your license key. Paste the key you get back into the field and click **Activate** — the Download tab's button unlocks immediately. A key only works on the machine it was issued for.

Downloading a profile

1. Connect the IRUSB cable to the meter's IR window and to your PC.
2. On the meter, press **SHIFT + RECALL** so the display shows "**Ir SEnd**". The app can only talk to the meter while it's in this mode.
3. In the app, open the **Download** tab.
4. Fill in **Plant** and **Tunnel** — free text, so use whatever naming you already use on the floor (a tunnel doesn't have to be numbered the same way at every plant).
5. Under **Meter Showed (at log time)**, select whether the meter's display was set to °C or °F when the log was recorded. This is just a record of what you saw on the meter — the app always stores and can display the data in either unit regardless.
6. Click **Download Profile**. The status line will say "Downloading..." while it works.



The Download tab. The button stays disabled, with an explanation, until a license is activated.

If the download fails: double check the meter is actually showing "Ir SEnd" (it times out and drops back to the normal display after a while, so you may need to press SHIFT + RECALL again), and that the cable is seated in the meter's IR window, not just resting near it.

Once a download succeeds, the app switches to the **Chart** tab automatically and plots what you just pulled off the meter.

If the meter was started and stopped more than once

If you logged more than one run on the meter before connecting to download (e.g. Tunnel 8, then Tunnel 9, without downloading or clearing memory in between), the meter hands back all of them in a single download. The app detects this automatically and asks for a tunnel name for each one before saving them as separate runs:

This download contains 2 separate logged runs -- the meter was started and stopped more than once before this download. Give each one a tunnel name:

RUN 1 — 61 SAMPLES, 20.6–35.4°C

Tunnel 8|

RUN 2 — 61 SAMPLES, 25.2–36.5°C

e.g. 3 or Superba A

OK Cancel

Shown automatically when a download contains more than one logged run.

Cancelling this dialog saves nothing from that download. Each run shown lists its sample count and temperature range to help you tell them apart.

Viewing and comparing charts



The Chart tab, two runs overlaid. Hovering shows a dot on each curve at the current time, and its live value in the legend.

Plotting more than one run

Double-click any entry in **Recent Profiles** to add it to the graph, in its own color. Double-click it again to remove just that one. This lets you overlay several runs to compare them directly — the highlighted (bold, colored) entries in the list are the ones currently on the graph.

Lining up runs that started at different times

The meter's logging start time is set by hand, so two runs of the same tunnel rarely start at exactly the same point in the process. To fix that: **click and drag directly on a curve** to slide it left or right in time (temperature is left alone) until its start and end line up with the run you're comparing it to. A small label shows the run name and current time offset while you drag.

Double-click a curve to reset its offset back to zero. Each run remembers its offset for as long as it stays on the graph, even if you switch °C/°F or temporarily remove and re-add it from Recent Profiles.

Reading the graph

- Move the mouse over the graph — a vertical line marks the time, a dot appears on every plotted curve at that time, and the legend updates to show each curve's temperature there, alongside the time itself.
- Scroll the mouse wheel to zoom the time axis in or out.
- Hold **Ctrl** and scroll to zoom the temperature axis instead (it's capped to a fixed 0–195°C

/ 0–390°F range so the scale stays meaningful).

Exporting what's on the graph

Button	What it does
Export Graph as PNG	Saves an image of the graph exactly as it looks right now
Export Plotted Data (CSV)	Saves the time/temperature numbers for every run currently plotted, including any drag alignment you've applied
Export Plotted Data (Excel)	Same data, in a spreadsheet with a built-in chart already drawn
Clear Graph	Removes everything from the graph (doesn't delete anything from the database)

Searching past profiles

The screenshot shows the 'Superba Tunnel Profiler' application with the 'Database' tab selected. The interface includes search filters for 'Plant' and 'Tunnel', date range selectors for 'From' (7/6/2026) and 'To' (8/6/2026), and 'Search' and 'Clear' buttons. Below the filters is a table with the following data:

Date	Plant	Tunnel	Peak	Points	Logged
2026-08-06T16:33:30+00:00	Bridgeport	Superba 8 - Run 1	35.4°C	61	°C
2026-08-06T16:33:30+00:00	Bridgeport	Superba 8 - Run 2	36.5°C	61	°C
2026-08-06T15:07:55+00:00	Bridgeport	Superba 8	35.4°C	61	°C
2026-08-05T18:38:23+00:00		Superba A	92.7°C	31	°C
2026-08-05T18:07:02+00:00		Superba A	92.7°C	31	°C
2026-08-05T18:03:46+00:00		Superba A	36.9°C	39	°C
2026-08-05T17:46:33+00:00		Superba A	28.7°C	39	°C
2026-08-05T17:44:32+00:00		Superba A	27.4°C	31	°C
2026-08-05T17:41:58+00:00		Superba A	27.4°C	31	°C

At the bottom of the table, it says '9 RUNS'. To the right of the table are buttons for 'Export Data (CSV)' and 'Print Report'. The status bar at the bottom left shows 'Idle'.

The Database tab: every profile ever downloaded, searchable.

Use this tab once you have more history than fits in the Chart tab's Recent Profiles list.

- Filter by **Plant** and/or **Tunnel** (matches partial text), and by a date range using the **From** / **To** checkboxes.
- **Double-click** a row to open it on the Chart tab.
- **Right-click** a row for a menu to export just that run as a PNG, CSV, or Excel file.
- Select a row and use **Export Data (CSV)** or **Print Report** at the bottom for the same

options, plus a one-page printable summary with the run's stats and its graph.

Settings

Display Unit controls whether temperatures are shown in °C or °F throughout the app — on the graph, in both run lists, and in exports. It's independent of what the meter was showing when a profile was logged. See **Activating your license** above for the License card.

Where your data lives

Every downloaded profile is stored in **superba_profiler.db**, a single file that sits right next to **SuperbaTunnelProfiler.exe**. If you move or copy the app to a new folder or PC, take that file with it (or start a fresh one on purpose). There's no cloud sync — it's local to whichever folder the .exe is run from.

Troubleshooting

Problem	Try this
Download button stays disabled	Activate a license for this machine on the Settings tab -- see Activating your license above.
Meter port shows "Not detected"	Reconnect the IRUSB cable, and confirm Windows sees it as a COM port (Device Manager).
Download button does nothing / times out	The meter must be showing "Ir SEnd" (SHIFT + RECALL). It reverts on its own after a period of inactivity, so re-trigger it right before clicking Download.
"The meter returned zero readings"	The selected log on the meter is empty. Recall through the meter's stored logs to confirm there's actually data on it.
A run is missing from Recent Profiles	That list only shows the 50 most recent. Use the Database tab to search further back.
A dragged curve won't line up exactly	Zoom in on the time axis (scroll wheel) first — it's much easier to nudge a curve precisely at a higher zoom level.